

Quadratics & Nonlinear Equations



Solving quadratics by factoring

- 1 Solve $x^2 - 5x + 6 = 0$ by factoring.
 - 2 Solve $2x^2 + 7x - 4 = 0$ by factoring.
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Solving quadratics with the quadratic formula

- 3 Solve $x^2 + 4x - 1 = 0$ using the quadratic formula, giving exact answers.
 - 4 Use the discriminant to determine the number of real solutions of $3x^2 - 2x + 5 = 0$, without solving the equation.
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Completing the square and vertex form

- 5 Write $f(x) = x^2 - 6x + 11$ in vertex form by completing the square, and state the vertex.
 - 6 A parabola has vertex $(2, -5)$ and passes through $(4, -1)$. Find its equation in vertex form.
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Finding zeros, maximum, and minimum values

- 7 A ball's height is modeled by $h(t) = -16t^2 + 64t + 5$, where t is time in seconds. Find the maximum height and the time at which it occurs.
 - 8 Find the zeros of $f(x) = x^2 - 9$, and interpret them as x -intercepts on a graph.
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Systems with one linear and one quadratic equation

- 9 Solve the system: $y = x^2 - 2x - 3$ and $y = x - 1$.
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Exponential growth and decay

- 10 A population of bacteria triples every hour, starting with 200 bacteria. Write an exponential model and find the population after 4 hours.

- 11 A car worth 24000 dollars depreciates by 15 percent each year. Write an exponential model and find its value after 3 years, to the nearest dollar.
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Radical and rational equations

- 12 Solve $\sqrt{2x + 3} = 5$, and check your solution.
- 13 Solve $\frac{3}{x-2} = \frac{1}{x+1}$, and state any restrictions on x .
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Nonlinear relationships in context

- 14 The area of a square garden is modeled by $A(s) = s^2$, where s is the side length in meters. If the area must be at most 144 square meters, find the maximum possible side length.
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Factoring special products

- 15 Factor $9x^2 - 25$ completely.
- 16 Factor $x^2 + 10x + 25$ completely, and identify the special pattern used.
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Converting between standard, factored, and vertex forms

- 17 A parabola is given in factored form as $y = 2(x - 1)(x + 3)$. Convert this to standard form.
- 18 A parabola has zeros at $x = 2$ and $x = 6$, and passes through $(4, -8)$. Find its equation in factored form.
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Quadratic word problems: projectile motion

- 19 A rocket is launched with height modeled by $h(t) = -16t^2 + 96t$, where t is time in seconds. Find when the rocket hits the ground, and find its maximum height.
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Quadratic optimization in business contexts

- 20 A company's revenue is modeled by $R(x) = -5x^2 + 300x$, where x is the price per item in dollars. Find the price that maximizes revenue, and find the maximum revenue.

Determining the number of solutions graphically

- 21** A parabola $y = x^2 - 4x + 7$ and a line $y = 2x - 5$ are graphed together. Use the discriminant of the resulting equation to determine how many times the graphs intersect.